PULSE AI

Technical Documentation

Verified architecture, financial methods, implementation evidence and limitations.

Shenzhen International FinTech Competition 2026
International Track — Topic E: SME Finance Copilot

Evidence date: 20 October 2026. All example business records are synthetic. No performance or adoption metrics are assumed. Supabase/PostgreSQL is the team's shared deployment database; the evidence run uses isolated SQLite.

01 / Executive overview

PULSE AI is a decision-support financial copilot for SMEs. A deterministic financial engine transforms transactions, invoices and recurring obligations into liquidity analysis and a baseline forecast. A Financial Digital Twin recomputes that same forecast under explicit alternative assumptions. The copilot adds explanations and tool orchestration; human approval controls consequential actions.

The central design principle is separation: financial arithmetic belongs to deterministic code; language generation is optional; simulations do not overwrite financial records; action proposals do not execute themselves.

Verified synthetic example	Actual result
Current cash	EUR 7,020.00
First forecast shortfall	2026-11-07
Baseline lowest cash / days negative	EUR -2,308.00 / 7
Supplier terms +15 days: lowest / days negative	EUR 2,064.00 / 0

These are demonstration outputs, not a forecast for a real SME and not evidence of predictive accuracy.

02 / The SME problem and solution

A business can show a positive historical cash-flow average while still facing a near-term timing gap. Owners often lack a dedicated CFO, and spreadsheets do not automatically connect overdue collections, scheduled payroll, supplier bills and alternative interventions.

PULSE helps the owner ask four practical questions: What is my current position? When could cash fall below zero? Which assumption or obligation causes the dip? What changes if I act differently?

In the synthetic case, trailing 30-day net flow is EUR 2,920.00. Historical burn is zero, so the burn-based runway is undefined rather than infinite. The scheduled-flow forecast nevertheless identifies a shortfall in 18 days. This distinction is intentional.

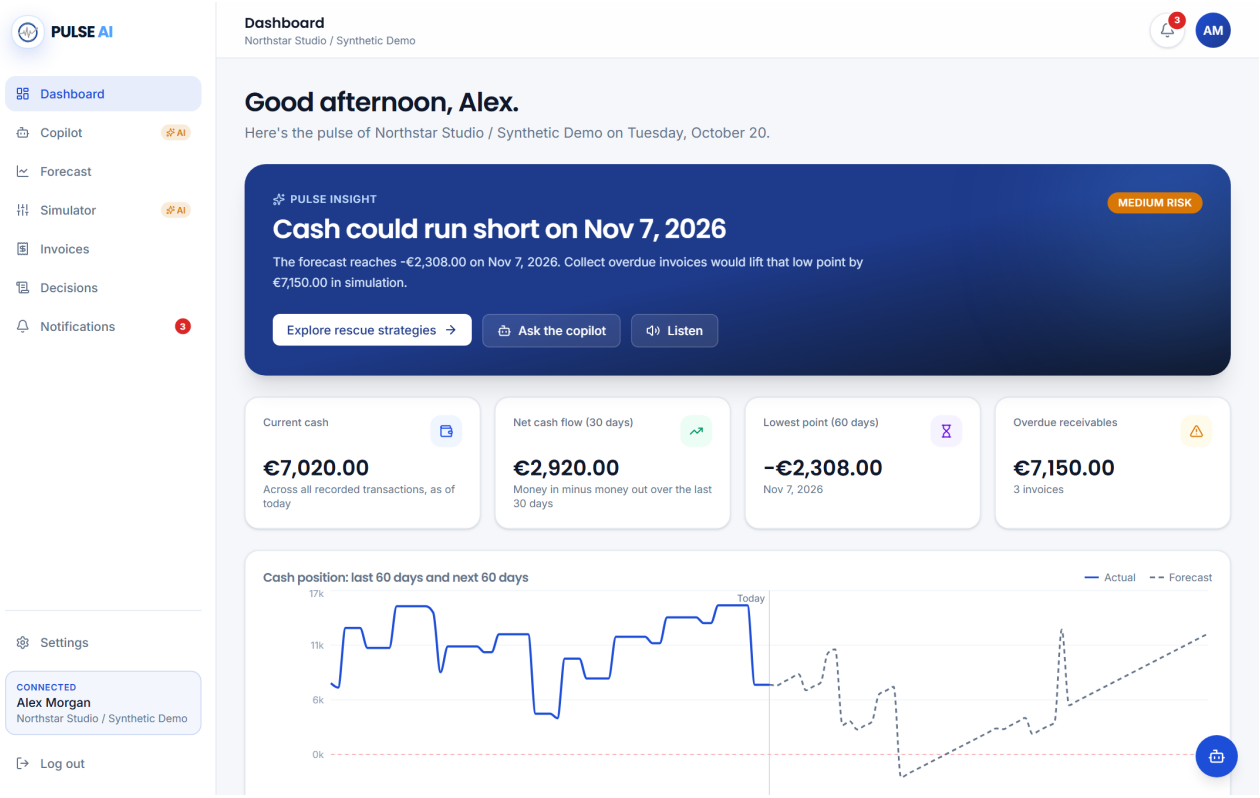


Figure 1. Real dashboard connected to the isolated synthetic demo backend.

03 / System architecture



Shared deployment architecture. Evidence backend: isolated SQLite with the same ORM and engines.

Browser requests carry a JWT bearer token. FastAPI resolves the authenticated user, requires account approval for business APIs, then checks organization membership and minimum role. The scoped financial input loader supplies records to the engines. Results are persisted as forecast runs, scenarios, proposals, conversations, notifications and audit records.

Supabase is used as the team's shared PostgreSQL deployment database, through SQLAlchemy's PostgreSQL connection. This is not a claim that Supabase Auth or PostgreSQL row-level-security policies are implemented. PULSE authentication and tenant checks remain application-level controls. The current local .env and submission fixtures use SQLite; a Supabase connectivity check is not evidenced here.

Email uses authenticated n8n webhooks and SMTP, with direct SMTP fallback for OTP. No bank or payment API is present. The submission action demonstration disables both outbound channels and honestly records QUEUED.

04 / Technology and data model

Layer	Actual implementation
Frontend	React 19, TypeScript, Vite, Tailwind CSS, React Query, Recharts, Framer Motion, Axios, React Router
Backend	FastAPI, Pydantic, SQLAlchemy, Alembic, Python Decimal
Database	Supabase/PostgreSQL shared deployment; SQLite isolated evidence fixture
Identity	bcrypt password hashing; JWT bearer tokens; HMAC-protected email OTP; platform account review
AI	OpenAI-compatible tool calling, Groq/OpenAI configuration, deterministic fallback
Automation	n8n 2.44.4 local harness, SMTP STARTTLS/implicit TLS, authenticated completion receipts
Import	CSV, Excel via openpyxl, supported Google Sheets URLs and column mapping

The ORM models users and memberships; organizations, business profiles and onboarding progress; transactions, invoices and recurring obligations; forecast runs/points; simulation scenarios/results; action proposals and recommendations; conversations/messages; notifications, audit logs, OTPs and account reviews. The complete column-level schema is included in execution_evidence/database_schema.json.

The requirements file is the reproducible dependency input. The captured runtime version manifest is separate and does not assume that installed packages exactly match every pinned requirement.

05 / Financial algorithms

All currency arithmetic uses Decimal and quantization to two decimal places. Current liquidity is opening cash plus recorded inflows minus recorded outflows through the as-of date. The opening balance is defined before the first recorded transaction. Onboarding adjusts the stored opening balance when the user supplies a current balance after importing history.

Trailing net flow = sum(inflows) − sum(outflows) over the requested inclusive dates. Monthly burn = max(outflows − inflows, 0) / (lookback_days / 30). Burn-based runway = floor(current cash / (monthly burn / 30)); it is null when burn is not positive.

Outstanding and overdue balances use invoice amount minus amount_paid, excluding paid invoices. Concentration is the largest customer's outstanding receivables divided by total outstanding receivables. Expense trend compares recent and preceding outflow windows.

Risk rules flag negative cash, burn-based runway below 30 or 60 days, overdue receivables, and customer concentration of at least 50%. These are transparent rules, not a trained credit-risk score.

Independent evidence replays raw demo transactions to obtain EUR 7,020.00, validates every one of the 90 daily forecast recurrences, checks the no-change simulation against baseline, and verifies the exact EUR 8,000 day-one inventory delta. Invoice snapshots remain unchanged after simulations and queued execution.

Limits: calculations do not perform foreign-exchange conversion or accounting reconciliation. The current liquidity loop does not explicitly filter transaction status. Inclusive comparison windows can share a boundary date; these definitions need review before accounting-grade use.

06 / Cash-flow forecast methodology

For each projected day: cash[d] = cash[d−1] + known[d] + uncertain[d]. Known flow includes remaining invoice balances scheduled on their due date and modeled recurring obligations. Monthly obligations match day-of-month; weekly obligations repeat every seven days.

Uncertain flow is a flat trailing 90-day daily net average after excluding categories represented by recurring obligations. It is included only when there are at least 30 historical transactions. Confidence labels are high through day 14, medium through day 45, then low; insufficient history yields low confidence throughout.

These confidence labels are heuristic categories, not calibrated probabilities or prediction intervals. There is no seasonality model, macroeconomic forecast or trained time-series model. Unpaid overdue invoices are not automatically assumed collectible in the baseline. Invoice cash receipts and historical sales averages can overlap, so double-counting beyond recurring-category exclusion remains a modeling concern.

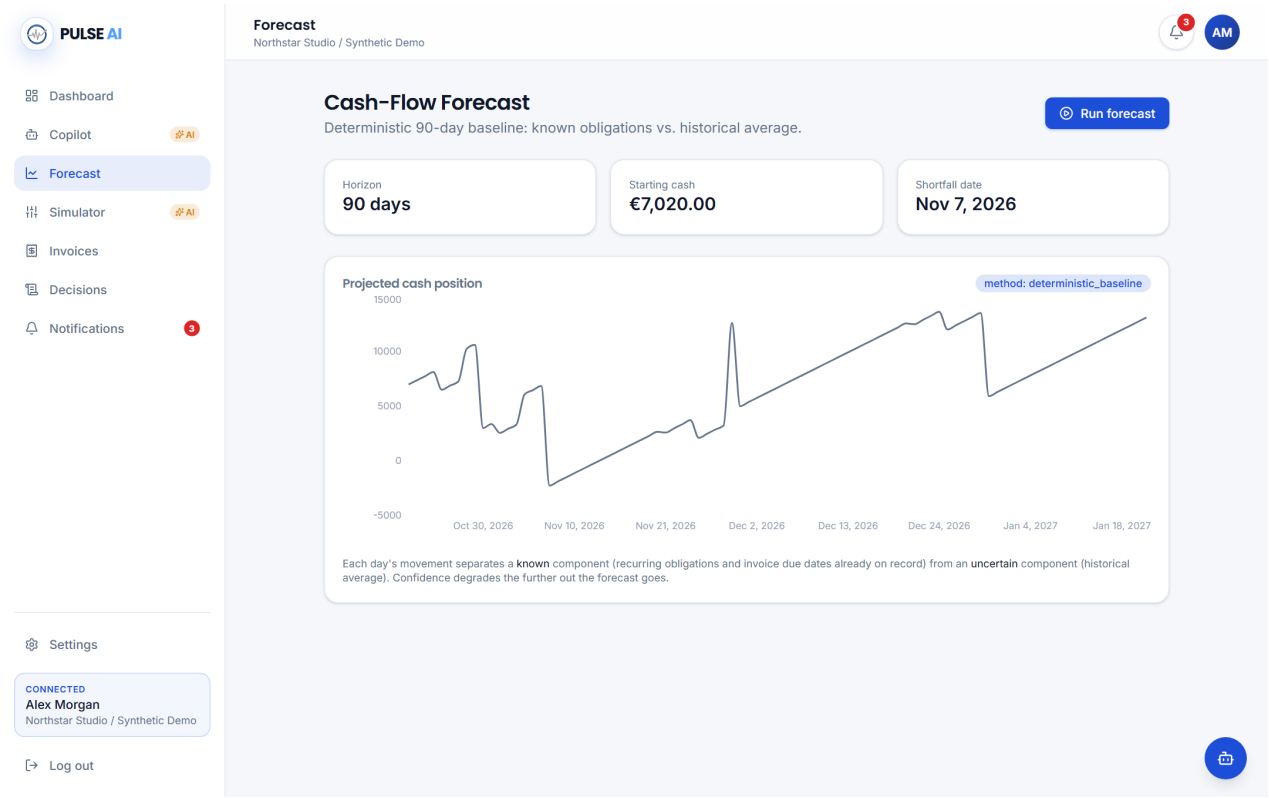


Figure 2. Actual 90-day forecast for the synthetic demo. First negative date: 2026-11-07

07 / Financial Digital Twin

Scenario assumptions operate on copied inputs. Supported changes include customer/supplier delays, revenue/expense shocks, one-time inventory purchases, financing proceeds/repayments and one-time cash injections. The internal tool layer additionally supports customer-specific delay and a stated overdue-collection assumption.

Both baseline and simulated paths use run_forecast. Comparison includes lowest cash point/date, days below zero and days until first shortfall. API impact_vs_baseline measures ending cash difference, not improvement of the minimum. A timing-only intervention may improve the cash minimum while leaving ending cash unchanged.

Supplier terms +15 days raises the actual simulated minimum by EUR 4,372.00, from EUR -2,308.00 to EUR 2,064.00. This removes negative-cash days in this particular synthetic scenario. It is conditional on the assumed extension and is not a guarantee of supplier consent.

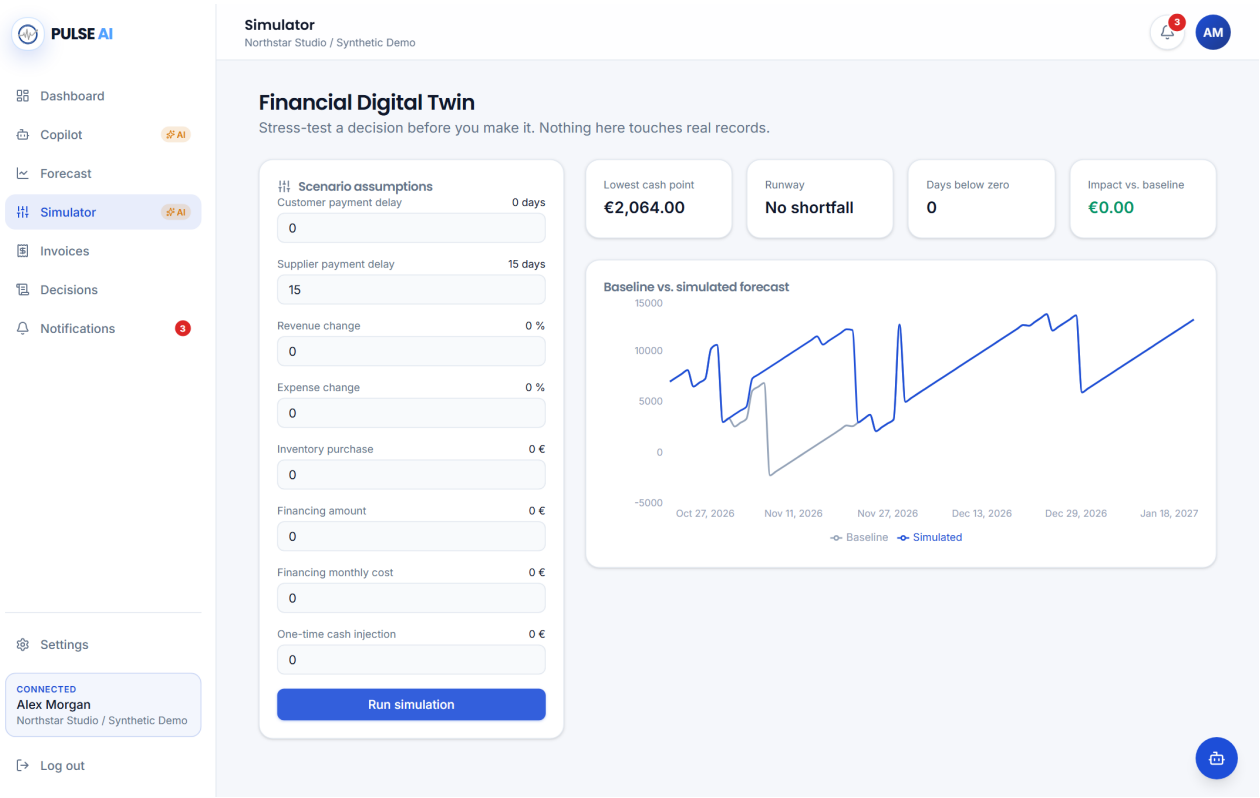


Figure 3. Real supplier-extension simulation and baseline comparison.

08 / Specialist agents and tools

Question + scoped context — Orchestrator: intent / tool loop — Financial / forecast / strategy tools — Explanation + proposal

Risk Guardian: role/scope checks and prompt constraints. Workflow execution remains outside the LLM loop.

The specialists are logical roles in one application, not separately deployed autonomous services. The tool registry includes get_financial_summary, forecast_cash, simulate_scenario, get_rescue_plans, list_invoices and propose_invoice_reminder. Tool results carry computed values and evidence.

With a provider key, the orchestrator permits up to six model/tool rounds and dispatches only registered tools. Unknown tools are rejected. Reminder drafting requires at least ACCOUNTANT. The model does not receive a direct execution tool. With no provider, keyword-based deterministic responses and traces remain available.

The demo and screenshots use deterministic mode. LLM tool-loop behavior is covered by repository tests with mocked provider responses, not by an actual external model inference in this evidence run. Generated language is not completely mechanically verified against tool outputs; avoid claiming that every possible LLM statement is guaranteed correct.

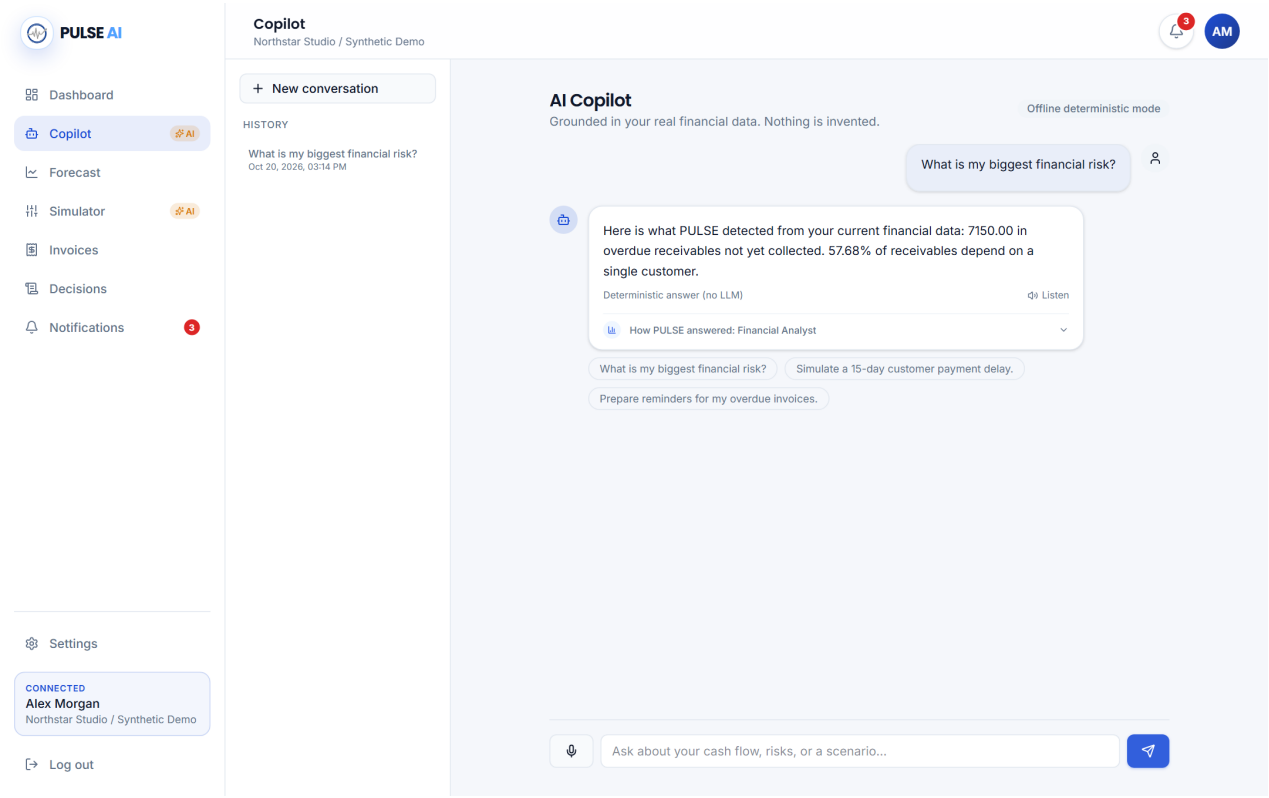


Figure 4. Genuine deterministic copilot response and trace.

09 / Retrieval and data grounding

RAG is not implemented. There is no embedding pipeline, vector database or semantic document retrieval index. Grounding is through structured financial input loading and registered deterministic tools.

The Excel/CSV/Google Sheets importer is a tabular ingestion feature, not retrieval-augmented generation. It offers column mapping, preview and duplicate detection. File size and row limits exist, but streaming upload limits, compressed workbook resource limits and adversarial input handling require hardening.

For a future RAG extension, keep document retrieval read-only, enforce tenant scope before retrieval, attach provenance and distinguish retrieved claims from engine-computed figures. This is a roadmap item, not submission functionality.

10 / Automation and human approval

The five n8n exports cover signup verification, cash-risk email, a financial digest, approved reminder dispatch and a recovery callback. Header-authenticated webhooks consume the request body and return explicit completion receipts. The local digest copy is on demand, while the source export also contains a schedule. Enabling recurring delivery requires explicit operational configuration.

PULSE creates action proposals, requires an authorized human approval, then checks approval and organization scope again at execution. The recovery workflow calls this guarded endpoint instead of bypassing it. SENT requires a confirmed submission; absence of a configured dispatcher yields QUEUED.

In this package's demo, no customer email is sent. The recorded sequence proves HTTP 409 before approval, UI approval, QUEUED after execution, and HTTP 409 on sequential replay. An earlier local n8n integration run confirmed all five workflows with real SMTP; its sanitized receipt summary is included separately with its provenance.

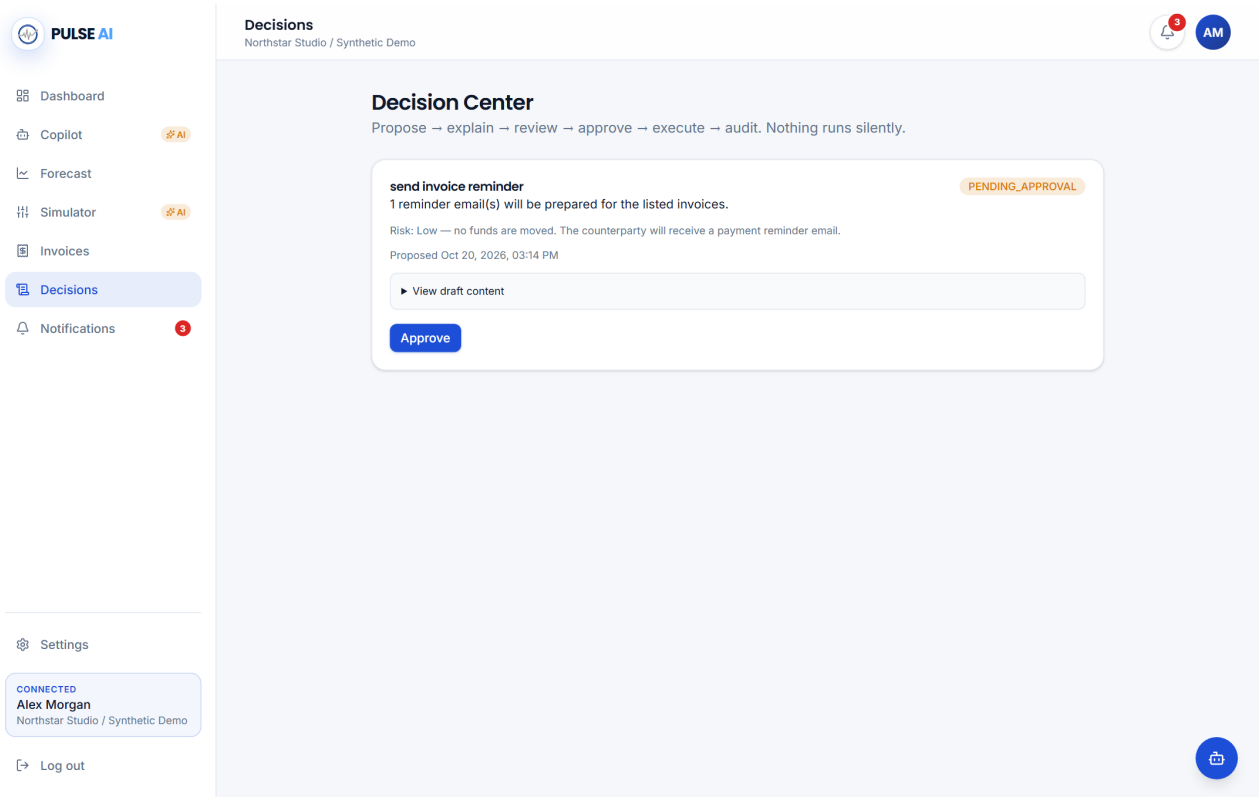


Figure 5. Actual pending approval screen in the synthetic workspace.

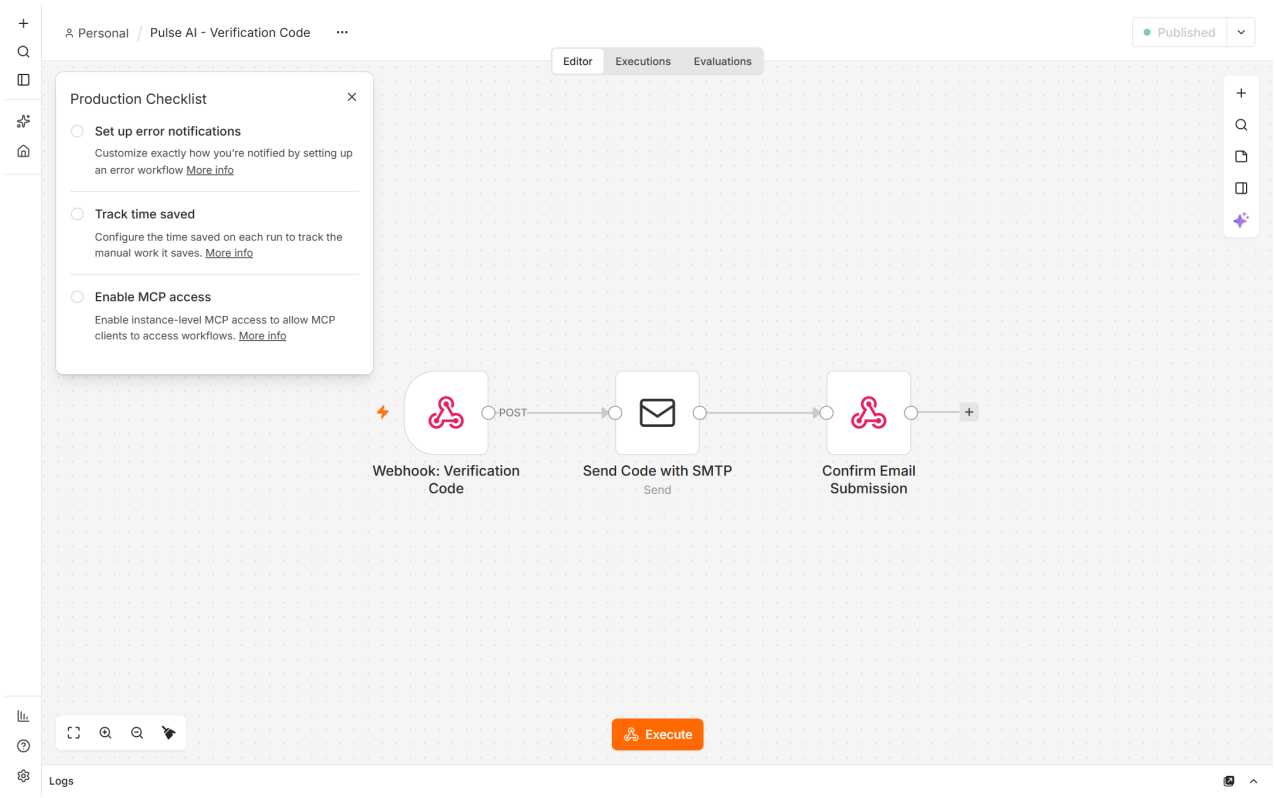


Figure 6. Real authenticated n8n editor, if captured; no credential fields are exposed.

11 / Authentication, tenancy and audit

New signup verifies an email OTP, stores a bcrypt password hash and creates a PENDING account. Platform approval is a separate privilege from organization ownership. Pending/rejected/suspended accounts can read their status but cannot use business APIs. Account-review updates compare the expected current status and record a review event atomically with the change.

Organization endpoints check membership. Protected operations require minimum roles. Conversation lookup additionally binds the conversation to both the organization and requesting user. Administrative access does not automatically grant access to tenant finances.

OTP codes are randomly generated, HMAC-hashed, short-lived, single-use, and have attempt/resend limits. Failed delivery removes an unusable code. SMTP supports STARTTLS and implicit TLS. These protections do not replace global rate limiting, deployment TLS or operational monitoring.

The browser stores bearer tokens in localStorage, exposing them to any successful XSS. Audit tables are application history, not cryptographically immutable evidence. For full threat modeling, priority fixes and control status, see the separate security assessment.

12 / Verification and reproduction

Fresh backend verification: 86 tests passed; the raw terminal output is provided. The frontend production build exited zero; it reports a large-chunk warning, not a functional failure. The automated evidence browser recorded no JavaScript page errors in the exercised routes.

Financial validation assertions and raw API request/response pairs are included. Twelve genuine screenshots and a real Chromium recording document the available pages. No network responses were mocked for these submission screenshots. All records shown belong to the isolated synthetic demo.

Reproduce: install backend requirements and frontend lockfile dependencies; run submission/scripts/demo_environment.py; start its serve mode on port 8002; run Vite on port 5175 with VITE_API_BASE_URL=http://127.0.0.1:8002; run capture_evidence.py. Its private state is generated under ignored .runtime and is excluded from the ZIP.

Shared deployment: every user's frontend must target the same backend, and that backend must use the same Supabase DATABASE_URL. Merely copying source code to two laptops does not share account state. Apply migrations to the intended shared database and configure SMTP/n8n on that backend.

The source commit and archive hash are documented separately. Uploading or generating this package does not constitute official competition submission.